Assignment 2

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Preparation

```
# packages
library("brms")
library("ggplot2")
library("tidyverse")

# data
load("data/income.RData")
```

data: `income.RData`

```
head(income)
```

```
##    ls emotst inc_hh emotst_cat
## 1 3.8    6.0      7       high
## 2 1.0    3.0      6        low
## 3 5.0    4.0     10        low
## 4 4.2    3.5      7        low
## 5 4.4    6.0      2       high
## 6 3.8    6.5      7       high
```

The data set contains four variables

Variable	Description
`ls`	Life satisfaction (1-5)
`emotst`	Emotional stability (1-7)
`emotst_cat`	Emot. stability (high-low)
`inc_hh`	Household income (in 10.000)

Excercise 1

Fit the following Bayesian model to assess the effect of household income on life satisfaction.

```
# Casewise exclusion of missing data
income <- na.exclude(income)

# This function can be used to figure out the structure of the
# priors for the sepcific model including names of the predictors
# default_prior(ls ~ inc_hh,
#               data = income)

bprior <- c(brms::prior(normal(3, 2), class = Intercept)
            , brms::prior(normal(0, 1), class = b, coef = inc_hh)
            , brms::prior(normal(0, 2.5), class = sigma))
```

```r
model.1 <- brm(ls ~ 1 + inc_hh
               , data = income
               , prior = bprior
               , silent = 2
               , refresh = 0)
```

```
## Running /usr/lib/R/bin/R CMD SHLIB foo.c
## using C compiler: ‘gcc (Ubuntu 15.2.0-4ubuntu4) 15.2.0’
## gcc -std=gnu23 -I"/usr/share/R/include" -DNDEBUG   -I"/home/julia/R/x86_64-pc-linux-gnu-library/4.5/R
## In file included from /home/julia/R/x86_64-pc-linux-gnu-library/4.5/RcppEigen/include/Eigen/Core:19,
##                  from /home/julia/R/x86_64-pc-linux-gnu-library/4.5/RcppEigen/include/Eigen/Dense:1,
##                  from /home/julia/R/x86_64-pc-linux-gnu-library/4.5/StanHeaders/include/stan/math/pri
##                  from <command-line>:
## /home/julia/R/x86_64-pc-linux-gnu-library/4.5/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:10:
##   679 | #include <cmath>
##       |          ^~~~~~~
## compilation terminated.
## make: *** [/usr/lib/R/etc/Makeconf:202: foo.o] Error 1
```

a) Consider the priors. Do you think they are a good choice? Run a prior prediction and change the priors if necessary.

b) Fit the model with the adjusted priors. Ensure convergence and visualize the posterior distributions of all parameters. Interpret the results.

Solution

- Y_i : Life satisfaction of the ith person
- x_i : Income

$$Y_i \sim \text{Normal}(\mu_i, \sigma^2)$$
$$\mu_i = \beta_0 + x_i\beta_1$$

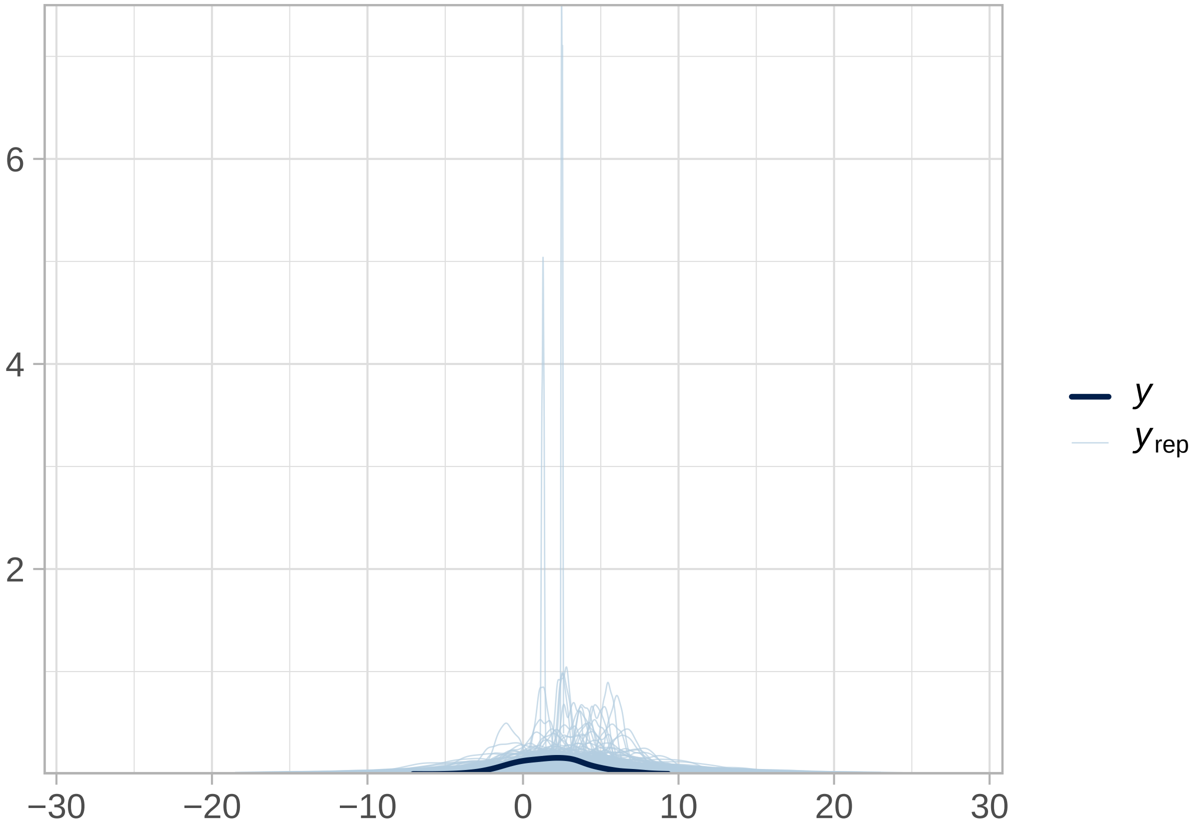
1a. The `ls`-values are between 1 and 5, so effects cannot be larger than one step on the life satisfaction scale. The mean is probably around 3. Sigma, the noise standard deviation can also not be bigger than 2-3 because of the range restriction of the data.

```r
dat <- data.frame(ls = rnorm(nrow(income), 2.5, 1)
                  , inc_hh = sample(1:12
                                    , size = nrow(income)
                                    , replace = T))
output <- capture.output(
model.prior.pred <- brm(ls ~ 1 + inc_hh
               , data = dat
               , prior = bprior
               , sample_prior = "only"
               , silent = 2
               , refresh = 0
               , cores = 2)
)
```

```
## Trying to compile a simple C file
```

```r
prior.predictions <- posterior_predict(model.prior.pred)
```

```r
bayesplot::ppc_dens_overlay(prior.predictions[1,], prior.predictions[1:500,], trim = T) +
  # xlim(-5, 10) +
  theme_light(base_size = 16)
```



Life satisfaction is on a scale from 1 to 5, so the range should be smaller.

1b.

```r
cprior <- c(brms::prior(normal(3, 1), class = Intercept)
            , brms::prior(normal(0, .5), class = b, coef = inc_hh)
            , brms::prior(normal(0, 1), class = sigma))
model.2 <- brm(ls ~ 1 + inc_hh
               , data = income
               , prior = cprior
               , cores = 4)
```

```
## Compiling Stan program...

## Trying to compile a simple C file

## Running /usr/lib/R/bin/R CMD SHLIB foo.c
## using C compiler: ‘gcc (Ubuntu 15.2.0-4ubuntu4) 15.2.0’
## gcc -std=gnu23 -I"/usr/share/R/include" -DNDEBUG   -I"/home/julia/R/x86_64-pc-linux-gnu-library/4.5/
## In file included from /home/julia/R/x86_64-pc-linux-gnu-library/4.5/RcppEigen/include/Eigen/Core:19,
##                  from /home/julia/R/x86_64-pc-linux-gnu-library/4.5/RcppEigen/include/Eigen/Dense:1,
##                  from /home/julia/R/x86_64-pc-linux-gnu-library/4.5/StanHeaders/include/stan/math/pr
##                  from <command-line>:
## /home/julia/R/x86_64-pc-linux-gnu-library/4.5/RcppEigen/include/Eigen/src/Core/util/Macros.h:679:10:
##   679 | #include <cmath>
##       |          ^~~~~~~
```

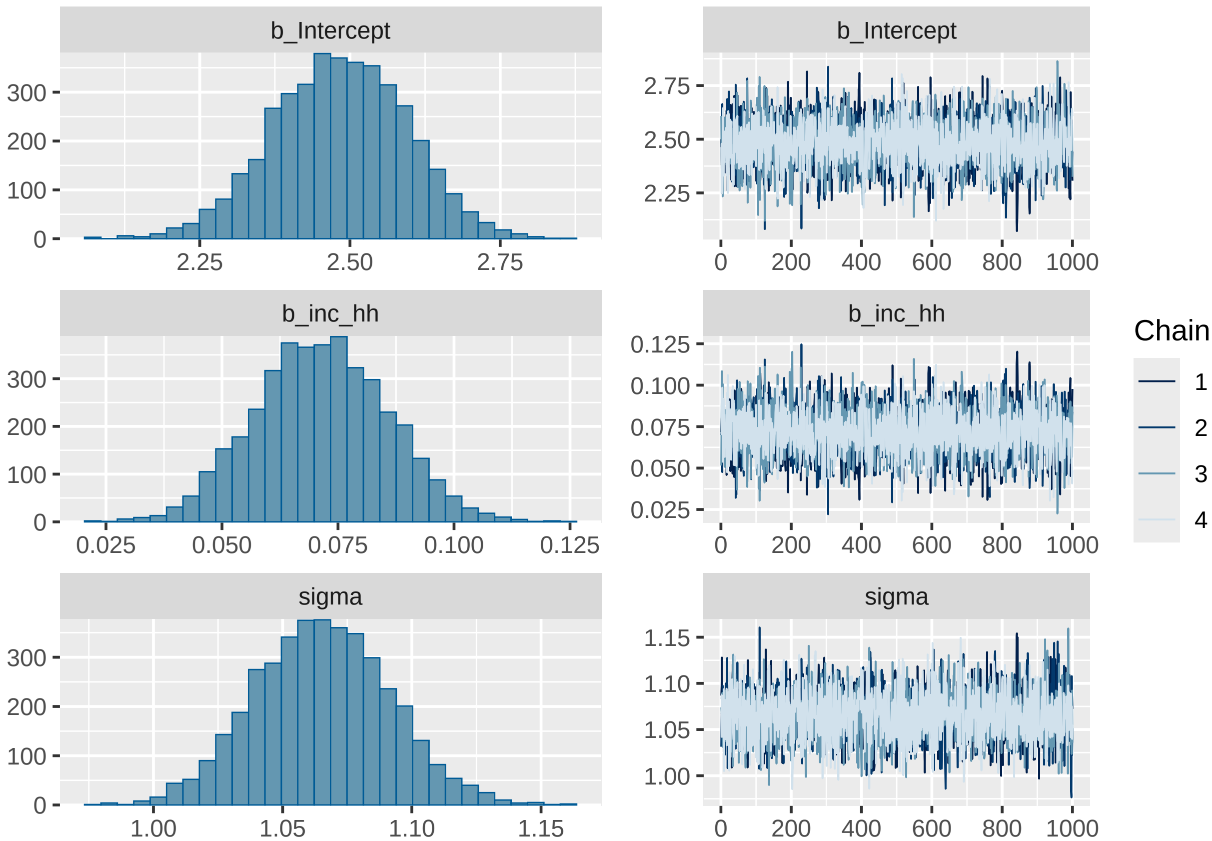
```
## compilation terminated.
## make: *** [/usr/lib/R/etc/Makeconf:202: foo.o] Error 1
```

```
## Start sampling
```

```r
model.2
```

```
##  Family: gaussian
##   Links: mu = identity
## Formula: ls ~ 1 + inc_hh
##    Data: income (Number of observations: 828)
##   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
##
## Regression Coefficients:
##           Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept     2.48      0.11     2.26     2.69 1.00     3909     2425
## inc_hh        0.07      0.01     0.04     0.10 1.00     4044     2777
##
## Further Distributional Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma     1.07      0.03     1.01     1.12 1.00     4138     2670
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
##       0.095 seconds (Total)
## Chain 3:
## Chain 4: Iteration:  800 / 2000 [ 40%]  (Warmup)
## Chain 4: Iteration: 1000 / 2000 [ 50%]  (Warmup)
## Chain 4: Iteration: 1001 / 2000 [ 50%]  (Sampling)
## Chain 4: Iteration: 1200 / 2000 [ 60%]  (Sampling)
## Chain 4: Iteration: 1400 / 2000 [ 70%]  (Sampling)
## Chain 4: Iteration: 1600 / 2000 [ 80%]  (Sampling)
## Chain 4: Iteration: 1800 / 2000 [ 90%]  (Sampling)
## Chain 4: Iteration: 2000 / 2000 [100%]  (Sampling)
## Chain 4:
## Chain 4:  Elapsed Time: 0.048 seconds (Warm-up)
## Chain 4:                0.042 seconds (Sampling)
## Chain 4:                0.09 seconds (Total)
## Chain 4:
```

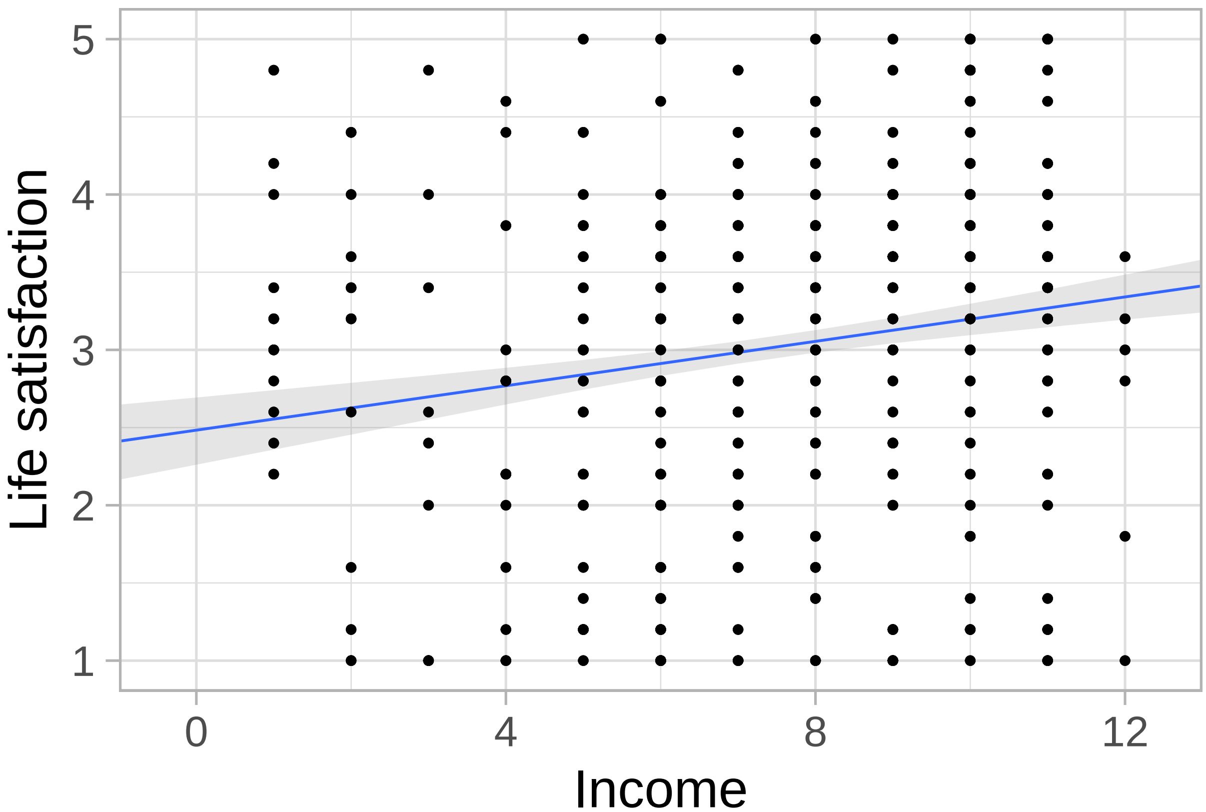
```r
plot(model.2)
```



Convergence is good. $\hat{R}$ and ESS are in the range and the caterpillars are hairy.

None of the posterior credible intervals includes 0. For an increase of 10.000 Euros of household income there is an increase of life satsifaction of about 0.07 points.

```
## Neue Daten werden erstellt, um die Regressionsgerade zu zeichnen
nd <- tibble(inc_hh = -1:13, 2)
## Regression wird auf die neuen Daten angewendet
f <- fitted(model.2, newdata = nd) %>%
  as_tibble() %>%
  bind_cols(nd)
income %>%
  ggplot(aes(x = inc_hh)) +
  geom_smooth(data = f,
              aes(y = Estimate, ymin = Q2.5, ymax = Q97.5),
              stat = "identity",
              alpha = 1/4, linewidth = 1/2) +
  geom_point(aes(y = ls),
             size = 2/3) +
  scale_x_continuous("Income", expand = c(0, 0)) +
  ylab("Life satisfaction") +
  theme_light(base_size = 20, )
```



c) Conduct a sensitivity analysis by fitting the model using three different priors.

Solution 1c.

```
cprior_a <- c(brms::prior(normal(3, .5), class = Intercept)
            , brms::prior(normal(0, .5), class = b, coef = inc_hh)
            , brms::prior(normal(0, .5), class = sigma))
output <- capture.output(
model.2_a <- brm(ls ~ 1 + inc_hh
               , data = income
               , prior = cprior_a
               , silent = 2
               , refresh = 0
               , cores = 4)
)
```

```
## Trying to compile a simple C file
```

```
model.2_a
```

```
##  Family: gaussian
##   Links: mu = identity
## Formula: ls ~ 1 + inc_hh
##    Data: income (Number of observations: 828)
##   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
##
## Regression Coefficients:
##             Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept       2.49      0.11     2.25     2.71 1.00     4035     2781
```

```
## inc_hh        0.07      0.01     0.04     0.10 1.00     3880     2928
##
## Further Distributional Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma     1.06      0.03     1.01     1.11 1.00     3845     3089
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
cprior_b <- c(brms::prior(normal(3, 2), class = Intercept)
            , brms::prior(normal(0, .5), class = b, coef = inc_hh)
            , brms::prior(normal(0, 2), class = sigma))
output <- capture.output(
model.2_b <- brm(ls ~ 1 + inc_hh
               , data = income
               , prior = cprior_b
               , silent = 2
               , refresh = 0
               , cores = 4)
)
```

```
## Trying to compile a simple C file
```

```
model.2_b
```

```
##  Family: gaussian
##   Links: mu = identity
## Formula: ls ~ 1 + inc_hh
##    Data: income (Number of observations: 828)
##   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
##
## Regression Coefficients:
##           Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept     2.48      0.11     2.25     2.71 1.00     3733     2484
## inc_hh        0.07      0.01     0.04     0.10 1.00     3771     2531
##
## Further Distributional Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma     1.07      0.03     1.01     1.12 1.00     4387     3115
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
cprior_c <- c(brms::prior(normal(3, 1), class = Intercept)
            , brms::prior(normal(-1, 1), class = b, coef = inc_hh)
            , brms::prior(normal(0, 1), class = sigma))
output <- capture.output(
model.2_c <- brm(ls ~ 1 + inc_hh
               , data = income
               , prior = cprior_c
               , silent = 2
               , refresh = 0
               , cores = 4)
```

```
)
```

```
## Trying to compile a simple C file
```

```
model.2_c
```

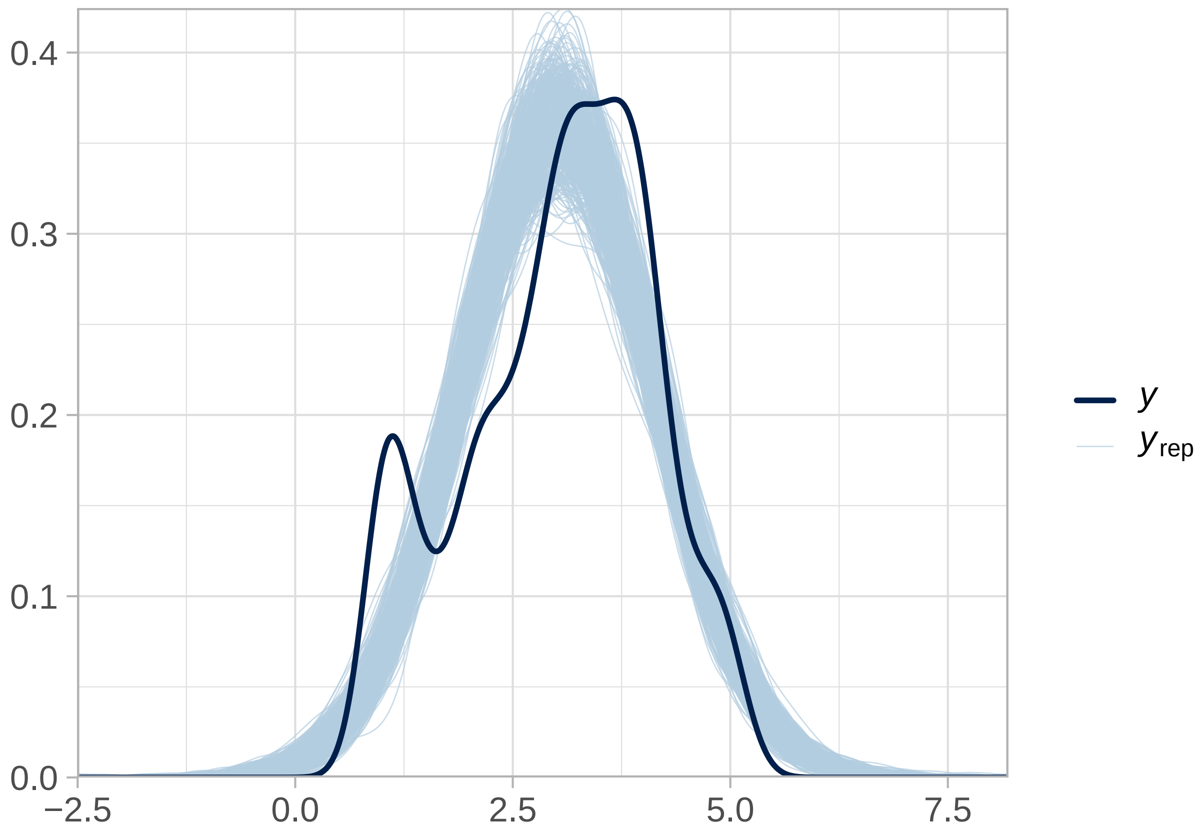
```
##  Family: gaussian
##   Links: mu = identity
## Formula: ls ~ 1 + inc_hh
##    Data: income (Number of observations: 828)
##   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
##
## Regression Coefficients:
##           Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept     2.48      0.11     2.26     2.71 1.00     4620     3003
## inc_hh        0.07      0.01     0.04     0.10 1.00     4537     2888
##
## Further Distributional Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma     1.07      0.03     1.01     1.12 1.00     4139     3192
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

d) Run a posterior prediction. Visualize the results.

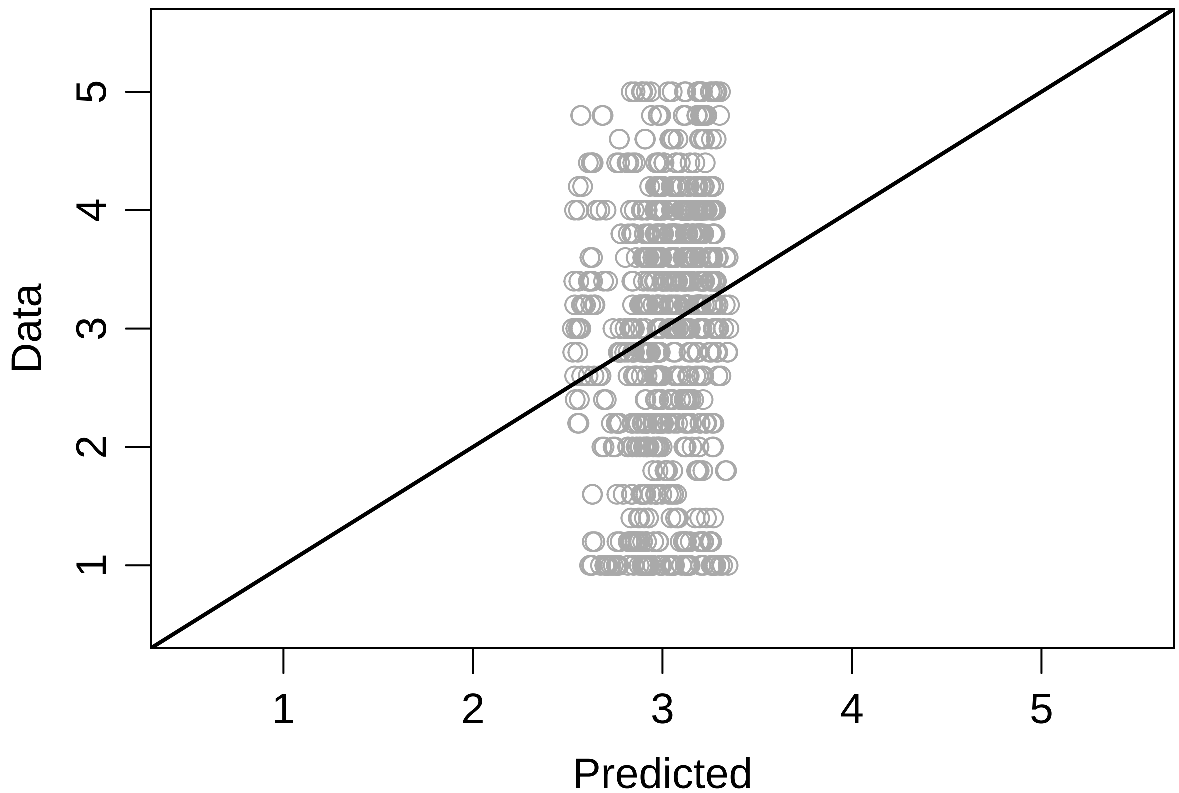
Solution 1d.

```
post.pred.1 <- posterior_predict(model.2)

bayesplot::ppc_dens_overlay(income$ls
                            , post.pred.1[1:500,]) +
  theme_light(base_size = 16)
```




```
par(mar = c(4,4,.5,.5), cex = 1.3, mgp = c(2, .7, 0))
post.pred.mean.1 <- colMeans(post.pred.1)
plot(post.pred.mean.1
     , income$ls
     , ylab = "Data", xlab = "Predicted"
     , pch = 1, col = "darkgray"
     , ylim = c(.5, 5.5), xlim = c(.5, 5.5))
abline(0, 1, lwd = 2)
```




```r
cor(post.pred.mean.1, income$ls)
```

```
## [1] 0.1769824
```